

A Micro-mobility Sensing System to Portray Riding Styles

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ABSTRACT

Riding style concerns the way a micro-mobility rider chooses to ride, and it plays a significant role in traffic safety. Portraying a rider's riding style is a useful way to guide them towards safer riding behaviors, and offer fine-grained information for insurance companies and bike-sharing companies to provide better services. To this end, we propose a micro-mobility sensing system to portray riding styles. Utilizing helmet-mounted and bicycle handle-mounted inertial sensors, our sensing system is able to monitor the micro-mobility movement status, detect the maneuver behaviors, and the safety check condition of riders. With 10 participants' experimental data, we present the feasibility of detecting maneuver behavior and recording the data that characterize the rider's riding style with our sensing system, and therefore make it a viable addition to the micro-mobility.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Micro-mobility, Wearable, Smartphone

ACM Reference Format:

Zengyi Han, Yuuki Nishiyama, and Kaoru Sezaki. 2022. A Micro-mobility Sensing System to Portray Riding Styles. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3544793.3560361>

1 INTRODUCTION

In recent years, sharing micro-mobility services have been promoted all over the world. As a result, bicycles and scooters are becoming an important mode of urban transportation and modern life. The corresponding reckless and aggressive riding behaviors also ensue, such as lane weaving [2], and single hand riding [1]. By understanding a rider's riding style and providing proper feedback, he may change riding styles to avoid the potential accident from happening. In vehicle driving style recognition, the inertial sensor is commonly used on the steering wheel to monitor the vehicle movement and maneuver of the driver [3]. However, the riding style not only includes the maneuver, but also the safety check level

of the rider before or during the maneuver. For example, failing to perform a safety check before turning also belongs to reckless riding behavior. Thus, we argue that head movement is also one important factor in riding style analysis.

To this end, we propose a micro-mobility sensing system for portraying riding style using two inertial sensors. Similar to the driving style recognition in the vehicle environment, we mounted one inertial sensor on the bicycle handle to monitor the movement and maneuver of micro-mobility. In order to track rider's visual safety check, we mounted the second inertial sensor on the rider's helmet: head movement is a strong indicator of the rider's eyesight, and the helmet, as an essential micro-mobility accessory, follows the same movement as the head of the rider. This makes the helmet an ideal device for unobtrusive monitor head movement.

2 SYSTEM DESIGN AND IMPLEMENTATION

2.1 System Design

The whole system setup is shown in Fig. 1. We leveraged the IMU sensor (contains 3-axis accelerometer and gyroscope) embedded in M5StickC, which is a portable (33 grams), open-source IoT development board (upper right in Fig. 1). We fixed it on the helmet to capture the rider's head movement with a 50 Hz sampling rate. While at the same time we mount an IMU-embedded smartphone on the bicycle's handlebar to monitor the rider's hand dynamics (maneuver). Two GoPro cameras were used to record participants' head movements and bicycle movements. The recorded video provided us with the ground truth for our subsequent analysis.



Figure 1: System equipments and setup.

We collected data from 10 participants from the university (7 male and 3 female, age 22 to 30). For safety reasons, we conducted the data collection inside the university campus (IRB approved). Each of the participants were asked to ride the bicycle along the

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UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
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ACM ISBN 978-1-4503-9423-9/22/09.
<https://doi.org/10.1145/3544793.3560361>

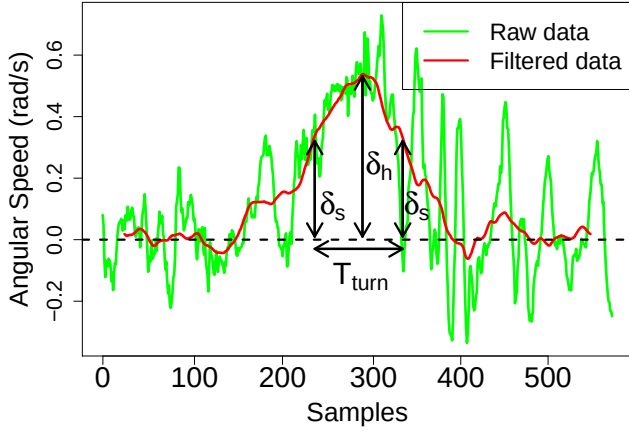


Figure 2: Example of a left turn in gyroscope reading.

campus for 5 laps. Four turns (all of them are 90°) in each lap, therefore, we have 200 turning examples ($10 \times 5 \times 4 = 200$).

2.2 Turning Detection and Head Movement Detection

Fig. 2 details an example of left turn in gyroscope reading. The raw data include noise, which caused by uneven road surfaces and the noise caused by pedaling, should be mitigated. A moving average filter was adopted to remove the noise from the raw data. For the filtered gyroscope data, we can easily observe that there is a peak that persists over a period of time. To this end, we define δ_s , δ_h , and T_{turn} three parameters for turning detection (Positive value reflects the angular speed towards the horizontal left, and vice versa). A turning is detected if the data meet the following three constraints: (1) all the gyroscope readings during a turn should be larger than δ_s , (2) the largest absolute value of the gyroscope reading during the turn should be no less than δ_h , and (3) the duration of a turn should be no less than T_{turn} .

From our road tests, we found that parameters of $\delta_s = 0.4$, $\delta_h = 0.6$ and $T_{turn} = 1$ second able to detect all the turns, and are thus used as default values for turn detection. Since head movement of riders normally appears prior to their maneuver, we use the head movement data before the maneuver happens.

3 EVALUATION

Fig. 3 shows the density distribution of δ_h and T_{turn} , respectively. Most of the participants have a maximal angular speed of around 1.2 rad/s . And the majority of the rider's turning time takes about 3 seconds. This number can help us detect the potential aggressive turning behaviors.

We plot participants' 10 seconds head movement (5 seconds before and 5 seconds after the left turning happens) in Fig. 4. As we can see that the head movements of each rider before the left turn are different. This is intuitively easy to understand: different road types, personal habits, and traffic conditions can cause differences in head movement. For example, P1, P2 and P3 checked both sides of the road, except P3 have an opposite head turning sequence with

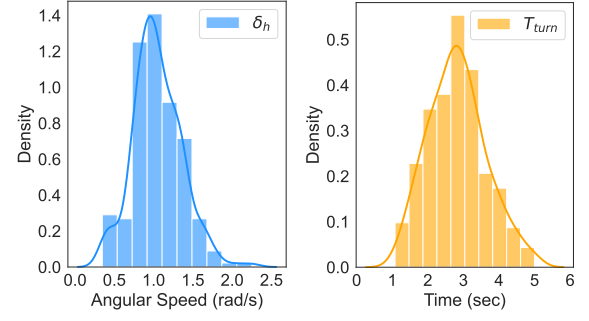


Figure 3: Density distribution of δ_h and T_{turn} .

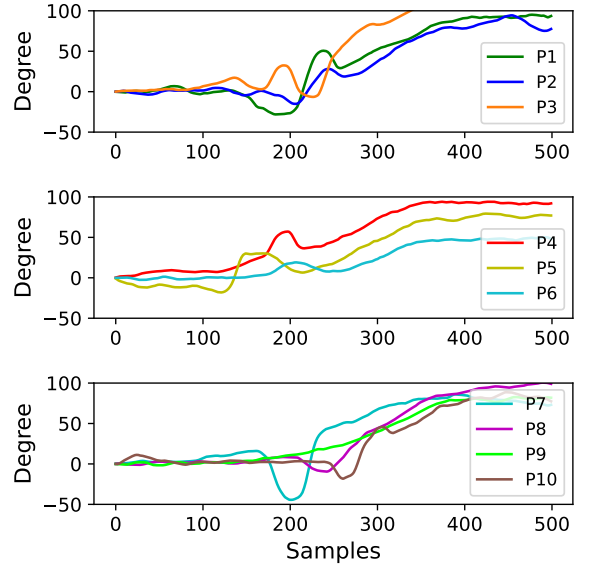


Figure 4: Examples of head movement before the turning.

the other two. P4, P5 and P6 only checked the left side of the road. P5's head movement prior to the other two. It is worth noting that P9 did not have any apparent head movement before the turning, which could be a potentially reckless riding behavior.

4 CONCLUSION

In this poster, a novel micro-mobility sensing system using multiple inertial sensors for portraying the rider's riding style is proposed. In contrast to existing solutions, we propose a sensing system that not only monitors the micro-mobility's maneuver but also the rider's head movement. We used 10 participants' data from their "left turn" maneuvers in our preliminary analysis. Our research shows that using helmet-mounted and bicycle handle-mounted inertial sensors can detect and record the data that characterize the rider's riding style, and therefore make it a viable addition to the micro-mobility. We envision our system can be used to provide riders with real-time feedback and generate rider profiles for insurance companies, and bicycle-sharing companies in the future.

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